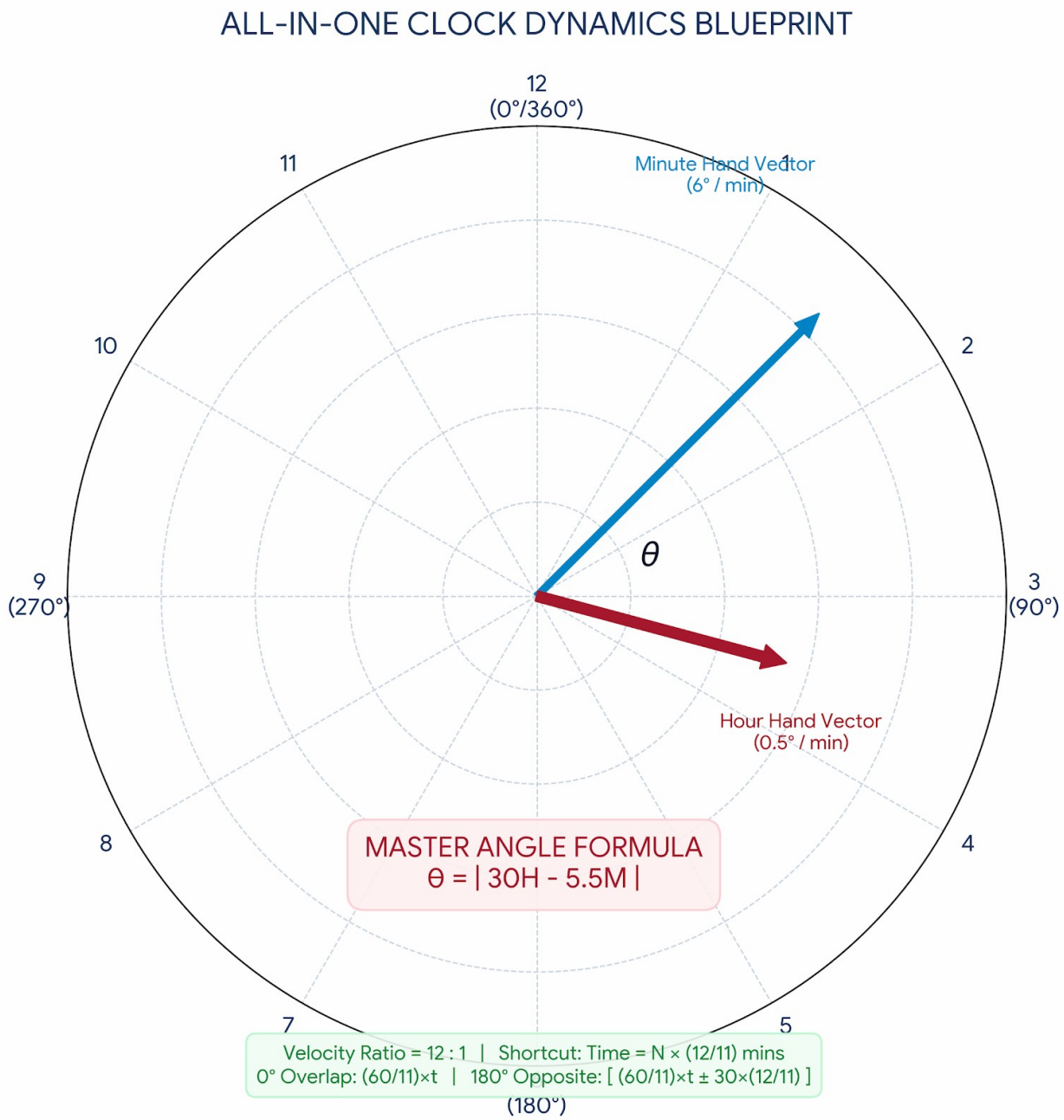
# Aptitude & Mental Ability Lecture Notes: Clock Problems






## 1. Fundamental Concepts of Clocks

The face or dial of a watch/clock is a circle whose circumference is divided into 60 equal parts, called minute spaces. A standard clock has two main hands: the smaller one is called the hour hand (or short hand), while the larger one is called the minute hand (or long hand).

- Angle Traced by Hour Hand in 12 hours = 360°
- Angle Traced by Minute Hand in 60 minutes = 360°
- Speed of Minute Hand: In 1 minute, the minute hand moves through (360° / 60) = 6° per minute.
- Speed of Hour Hand: In 1 minute, the hour hand moves through (360° / (12 × 60)) = 360° / 720 = 0.5° (or 1/2°) per minute.
- Relative Speed / Relative Gain: In 1 minute, the minute hand gains (6° - 0.5°) = 5.5° (or 5 ½°) over the hour hand.
- In 60 minutes, the minute hand gains 55 minute spaces over the hour hand.
- Coincidence: In every hour, both the hands coincide exactly once.
- The hands coincide 11 times in every 12 hours (between 11 and 1 o'clock there is only one common position at exactly 12 o'clock). Hence, the hands coincide 22 times in a single day (24 hours).
- Straight Line: The hands are in the same straight line when they are either coincident or opposite to each other.
- Opposite Directions: When the hands are in opposite directions, they are exactly 30 minute spaces apart (which corresponds to 180°).
- Right Angles: When the two hands are at right angles, they are exactly 15 minute spaces apart (which corresponds to 90°).
- The hands of a clock are at right angles twice in every hour, but in 12 hours they are at right angles 22 times (since there are common positions at 3 and 9 o'clock). Consequently, they are at right angles 44 times in a day.

## 2. Core Quantitative Formulae

**Formula A: Finding the Angle Between Hands at Time H:M**
If 'H' represents the hours and 'M' represents the minutes, the absolute angle θ between the hands can be computed directly using:

**ANGLE = |30H - (11/2)M|**

**Formula B: Exact Time of Overlapping/Coincidence between Hour t and t+1**
The exact time when the two hands overlap is given by:

**Time = (60/11) × t  minutes past t o'clock**

**Formula C: Exact Time of Right Angles (90°) between Hour t and t+1**
Since a 90° angle represents a 15-minute gap (or 15 min × 12/11 relative minutes), the direct formula is:

**Time = [ (60/11) × t ± 15 × (12/11) ] minutes past t o'clock**

**Formula D: Exact Time of Opposite Hands (180°) between Hour t and t+1**
Since a 180° angle represents a 30-minute separation, the direct formula is:

**Time = [ (60/11) × t ± 30 × (12/11) ] minutes past t o'clock**

## 3. Clock Faults: Gain and Loss of Time

Clocks that run too fast or too slow are common test topics. A standard correct clock has its hands coincide exactly every 65 5/11 minutes.

• True Time Interval Relation:

(True Time Interval) / (Time Interval in Incorrect Clock) = 1 / [1 ± (x / 60)]
Where (+) is used when the incorrect clock gains time, and (-) when it loses time.

• Total Time Gained or Lost Formula:

If both hands of an incorrect clock coincide at an interval of x minutes (where x ≠ 65 5/11), then:
Total Time Gained/Lost = 60 × T × [ (65 5/11 - x) / x ] minutes
(where T is the total hours elapsed on the correct time scale).

## 1. The Proportion of Speeds Between Clock Hands

📜 The Core Theory:

- **Minute Hand Speed:** Moves 360° in 60 minutes = 6° / minute
- **Hour Hand Speed:** Moves 360° in 12 hours (720 minutes) = 0.5° / minute
- **The Velocity Ratio:** Speed of Minute Hand / Speed of Hour Hand = 6° / 0.5° = 12 / 1

⭑ **Key Takeaway for Students:** The minute hand always covers angular distance exactly 12 times faster than the hour hand.

📝 Problem 1 (Speed & Displacement Tracking):

**Question:** A standard mechanical clock is started exactly at noon (12:00 PM). By 10 (25) minutes past 5:00 PM (7PM) on the same day, how many degrees has the hour hand turned?

Step-by-Step Solution:

• **Step 1: Calculate total elapsed time.**
From 12:00 PM to 5:10 PM = 5 hours + 10 minutes.
*Total minutes = (5 × 60) + 10* = **310 minutes.**

• **Step 2: Apply the individual hand speed.**
The hour hand moves at 0.5° per minute.
*Angular Movement = 310 minutes × 0.5°/min* = **155°.**

**Final Answer: The hour hand has rotated through 155°.**

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## 2. The Angles Between Several Positions of the Hands

📜 The Core Theory:

To calculate the absolute internal angle (θ) between the two hands at any given timestamp written as H hours : M minutes, use the Master Formula:

$$\theta = |30H - (11/2)M|$$

⚠ **Important Boundary Rule:** If the calculated angle θ turns out to be greater than 180°, it represents the exterior angle. Subtract it from 360° to get the normal inner acute/obtuse angle (360° - θ).

📝 Problem 2 (Reflex Angle Calculation):

**Question:** Find the reflex angle formed between the hands of a clock at exactly 10:25. (9.49)

Step-by-Step Solution:

• **Step 1: Identify your structural variables.**
Given time = 10:25 → H = 10, M = 25

• **Step 2: Plug the variables into the master formula.**
θ = |30(10) - (11/2)(25)|
*θ = |300 - 137.5|* = **162.5°**

• **Step 3: Convert to a Reflex Angle.**
A reflex angle must always be greater than 180°.
*Reflex Angle = 360° - 162.5°* = **197.5° (or 197 ½°)**

**Final Answer: The reflex angle at 10:25 is 197.5°.**

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## 3. Coincidental (0°) vs. Collinear & Opposite (180°) Positions

📿 The Core Theory:

• Coincidental Alignment (0° Shift):

Hands are directly overlapping on top of each other (0 minute spaces apart).

• Collinear & Opposite Alignment (180° Shift):

Hands form a perfectly straight line pointing away from each other (30 minute spaces apart).

⚡ **Shortcut Conversion Tool:** To find how long it takes to clear a specific number of 'minute spaces' gap, multiply that gap distance by the relative timing constant: **12/11**

📝 Problem 3 (Finding Coincidence / 0°):

**Question:** At what exact time between 1:00 PM and 2:00 PM will the hands of a clock coincide?

Step-by-Step Solution:

- **Step 1: Find the initial gap at the start of the hour.**
  At exactly 1:00 PM, the hour hand is at 1 and the minute hand is at 12.
  *This means the minute hand is lagging behind by exactly* **5 minute spaces.**

- **Step 2: Compute the time needed to close the gap.**
  To catch up, multiply the space deficit by the conversion constant (12/11):
  *Time = 5 spaces × 12/11 = 60/11 =* **5 5/11 minutes.**

  **Final Answer: The hands will coincide at 5 5/11 minutes past 1 o'clock.**

📝 Problem 4 (Finding Opposite Direction / 180°):

**Question:** At what exact time between 7:00 AM and 8:00 AM are the hands collinear but pointing in opposite directions?

Step-by-Step Solution:

- **Step 1: Analyze the target space setup.**
  *For hands to be opposite, they must maintain a strict distance of* **30 minute spaces** from each other.

- **Step 2: Check the baseline position at the hour mark.**
  At 7:00 AM, the hour hand is at 7 and the minute hand is at 12.
  *This represents an initial separation of* **35 minute spaces.**

- **Step 3: Calculate the movement needed.**
  *To shrink the initial 35-space gap down to a 30-space gap, the minute hand needs to close a net relative distance of: 35 - 30 =* **5 minute spaces.**

- **Step 4: Convert spaces to precise minutes.**
  *Time = 5 spaces × 12/11 = 60/11 =* **5 5/11 minutes.**

  **Final Answer: The hands become collinear and opposite at 5 5/11 minutes past 7 o'clock.**

## Examples :

## 4. Step-by-Step Solved Examples

**Example 1: Find the angle between the hour hand and the minute hand of a clock when the time is 3:25. Do for (8.27)**

*Solution (Method 1 - Standard Analytical Geometry):*

1. Angle traced by the hour hand in 12 hours = 360°.
2. Total hours at 3:25 = 3 + 25/60 hours = 3 + 5/12 = 41/12 hours.
3. Angle traced by the hour hand = (360° / 12) × (41/12) = 30° × (41/12) = 102.5° (or 102 ½°).
4. Angle traced by the minute hand in 60 min = 360°.
5. Angle traced by it in 25 min = (360° / 60) × 25 = 6° × 25 = 150°.
6. Required absolute angle = |150° - 102.5°| = 47.5° (or 47 ½°).

*Solution (Method 2 - Using Direct Angle Formula):*

Using ANGLE = |30H - (11/2)M| where H = 3 and M = 25:

ANGLE = |30(3) - (11/2)(25)| = |90 - 275/2| = |90 - 137.5| = |-47.5°| = 47.5°.

**Example 2: Determine the exact time between 6 and 7 o'clock when the two hands overlap with each other. Do for 3 clock and 8 o clock.**

*Solution:*

Using the coincidence formula where t = 6:

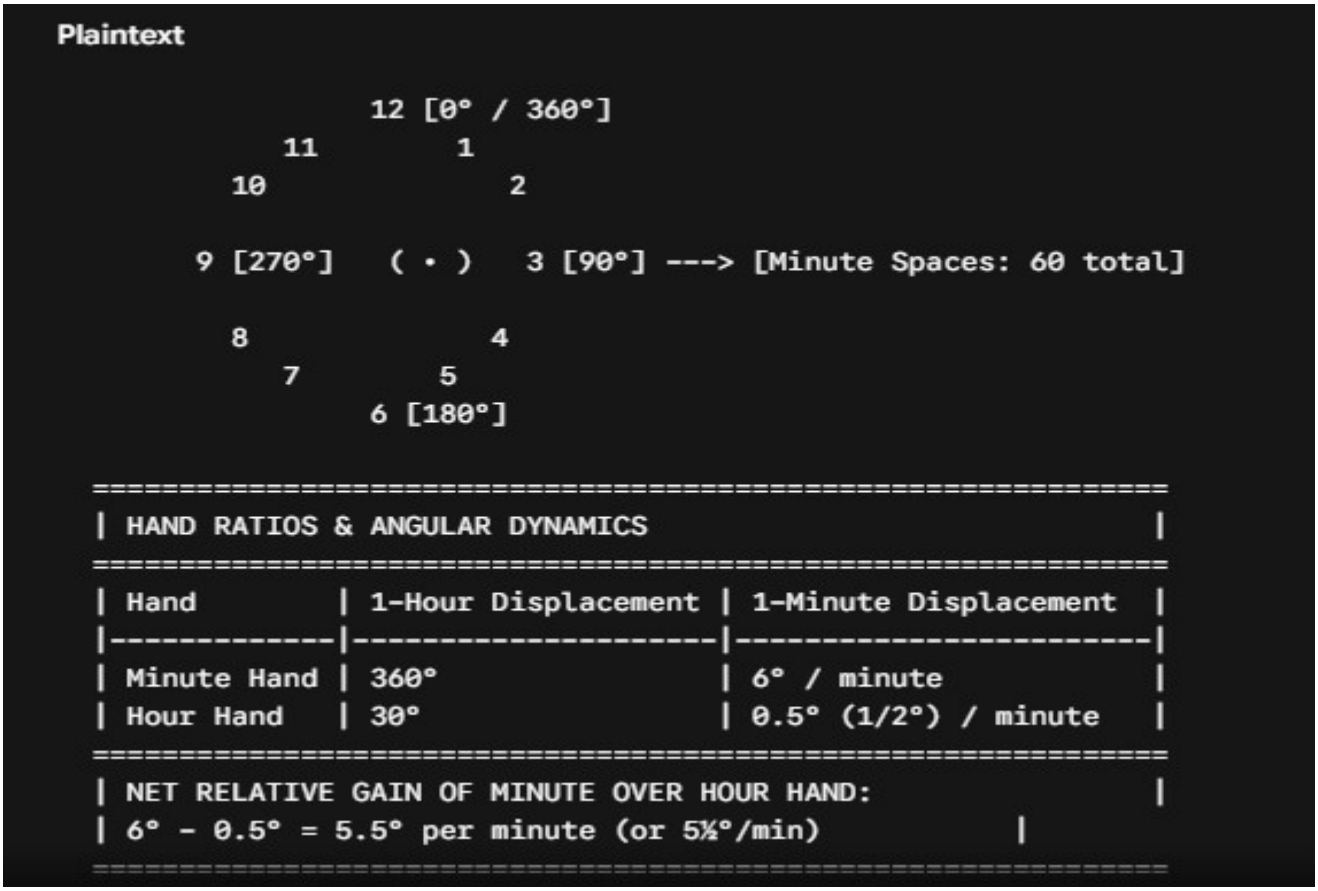
Time = (60/11) × 6 = 360/11 minutes past 6 o'clock.

Converting to mixed fraction: 360 ÷ 11 = 32 with a remainder of 8.

Therefore, the hands overlap at exactly 32 8/11 minutes past 6 o'clock.

Diagram :

Plaintext




| HAND RATIOS & ANGULAR DYNAMICS | | |
|---|---|---|
| Hand | 1-Hour Displacement | 1-Minute Displacement |
| Minute Hand | 360° | 6° / minute |
| Hour Hand | 30° | 0.5° (1/2°) / minute |
| NET RELATIVE GAIN OF MINUTE OVER HOUR HAND: | | |
| 6° - 0.5° = 5.5° per minute (or 5½°/min) | | |

## 14. If "H" is the hours and "M" is the minutes, then angle can be directly obtained by

**ANGLE = 30H − (11/2)M**

### Solution

We know that,

1. Hour hand moves ½ **degree in a minute** or **30 degrees in an hour**, and the minute hand travels **6 degrees in a minute**. Therefore, the minute hand gains **5.5 degrees per minute** over the hour hand.
2. When the time is **H : M (H hours : M minutes)**, the angle traversed by:
    - Hour hand = **30 × H**
    - Minute hand = **5.5 × M**

    (Considering the angle is measured with respect to the hour hand position from the standard 12:00 midnight position.)

3. Therefore, subtract the two quantities:

**Angle = 30H − 5.5M**

### Examples

1. **At 9:30**

Angle = 30 × 9 − (11/2) × 30

= 270 − 165

= **105°**

---

2. **At 12:00**

Angle = 30 × 12 − 0

= **360°**

---

3. **At 5:35**

Angle = 30 × 5 − (11/2) × 35

= 150 − 385/2

= **42.5°**

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## 15. The exact time between a given time interval (t, t + 1) when the two hands overlap with each other?

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## 16. The exact time between a given time interval (t, t + 1) when the two hands are at right angles?

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## 17. The exact time between a given time interval (t, t + 1) when the two hands are opposite to each other?

### Ex. 1

A watch which gains uniformly is **5 minutes slow at 8:00 a.m. on Sunday** and it is **5 minutes 48 seconds fast at 8:00 p.m. on the following Sunday**. When was it correct?

**Options:**

1. 6:00 p.m. on Wednesday
2. 7:20 p.m. on Wednesday
3. 6:00 p.m. on Tuesday
4. 7:20 p.m. on Tuesday

**Answer:** correct **3 days 11 hours 20 minutes after 8:00 a.m. on Sunday.**

**7:20 p.m. on Wednesday.**

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## Ex. 2

A clock is set right at **1:00 p.m.** If it gains **1 minute every hour**, what is the true time when the clock indicates **6:00 p.m.** on the same day?

### Options

1. 50 minutes past 5
2. 55 minutes past 5
3. 55 5/61 minutes past 5
4. 50 5/61 minutes past 5

**Answer: 55 5/61 minutes past 5**

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## Ex. 3

A clock gains **5 minutes in 24 hours**. It was set right at **10:00 a.m. on Monday**. What will be the true time when the clock indicates **10:30 a.m. on the next Sunday?**

### Options

1. 10:00 a.m. on Sunday
2. 9:00 a.m. on Sunday
3. 11:00 a.m. on Sunday
4. None of these

**Answer: 10:00 a.m. on Sunday**

## Example 8

**Question:**
A minute hand of a clock overtakes the hour hand at intervals of **63 minutes of correct time**.

How much time does the clock lose or gain per day?

### Options

1. **55 8/76 minutes**
2. **56 8/77 minutes**
3. **56 9/77 minutes**
4. **77 8/56 minutes**

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**65 5/11 − 63 = 2 5/11 minutes**

in every **63 minutes**.

---

**56 8/77 minutes**

## CLOCKS – DETAILED THEORY AND SOLVED EXAMPLES

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# 14. ANGLE BETWEEN THE TWO HANDS OF A CLOCK

## Formula

If:

H = Hour
M = Minutes

Then the angle between the two hands is:

ANGLE = |30H − (11/2)M|

### Why?

#### Step 1: Movement of the hour hand

The hour hand completes 360° in 12 hours.

Therefore,

Hour hand movement in 1 hour

= 360/12

= 30°

Hour hand movement in 1 minute

= 30/60

= 1/2°

So, at H:M,

Hour hand position = 30H + M/2

#### Step 2: Movement of the minute hand

The minute hand completes 360° in 60 minutes.

Therefore,

Minute hand movement in 1 minute

= 360/60

= 6°

So,

Minute hand position = 6M

#### Step 3: Difference between the two positions

Angle = Minute hand position − Hour hand position

= 6M − (30H + M/2)

= 6M − 30H − M/2

= (12M − M)/2 − 30H

= 11M/2 − 30H

Therefore,

ANGLE = |30H − 11M/2|

The smaller angle is normally taken if the question asks for "the angle between the hands."

---

### Example 1: Find the angle at 9:30

Here,

H = 9
M = 30

Angle

= |30 × 9 − (11/2) × 30|

= |270 − 165|

= 105°

Therefore,

ANSWER = 105°

---

### Example 2: Find the angle at 12:00

H = 12
M = 0

Angle

= |30 × 12 − (11/2) × 0|

= |360 − 0|

= 360°

But the hands are actually together at 12:00.

Therefore, the angle between them is:

ANSWER = 0°

Note: 360° represents a complete revolution, but the smaller angle between the hands is 0°.

---

### Example 3: Find the angle at 5:35

H = 5
M = 35

Angle

= |30 × 5 − (11/2) × 35|

= |150 − 385/2|

= |300/2 − 385/2|

= |-85/2|

= 85/2

= 42.5°

Therefore,

ANSWER = 42.5°

---

## 15. EXACT TIME WHEN THE TWO HANDS OVERLAP

We know:

Minute hand speed = 6° per minute

Hour hand speed = 1/2° per minute

Therefore, relative speed of minute hand over hour hand:

= 6 − 1/2

= 11/2° per minute

At exactly t:00, the hour hand is t hour-spaces ahead of the minute hand.

One hour-space = 30°.

Therefore, separation at t:00:

= 30t°

Time required to close this gap:

Time = Distance / Relative speed

= 30t / (11/2)

= 30t × 2/11

= 60t/11 minutes

Therefore,

EXACT TIME OF OVERLAP BETWEEN t AND t+1

= t hours + (60t/11) minutes

or

ANSWER = 60t/11 minutes past t.

---

## Example: Between 6 and 7

Here,

t = 6

Time required:

= 60 × 6 / 11

= 360/11

= 32 8/11 minutes

Therefore,

ANSWER = 6:32 8/11

---

## Example: Between 2 and 3

At 2:00, the two hands are 2 hour-spaces apart.

Each hour-space = 5 minute-spaces.

Therefore,

Gap = 2 × 5

= 10 minute-spaces.

The minute hand gains 55 minute-spaces in 60 minutes.

Therefore,

10 minute-spaces are gained in:

= (60/55) × 10

= 600/55

= 120/11

= 10 10/11 minutes

Therefore,

ANSWER = 2:10 10/11

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## 16. EXACT TIME WHEN THE TWO HANDS ARE AT RIGHT ANGLES

Two hands are at right angles when the angle between them is:

90°

We already know that the hands overlap at:

60t/11 minutes past t.

At the time of overlap, the angle is 0°.

To make a right angle, the minute hand has to move ahead by 90° relative to the hour hand.

Relative speed:

= 11/2° per minute

Therefore,

Time required for 90°:

= 90/(11/2)

= 90 × 2/11

= 180/11

= 16 4/11 minutes

Therefore, one right-angle time is:

60t/11 − 180/11

or

60t/11 + 180/11

depending on which occurrence lies inside the interval.

The direct formula is:

RIGHT ANGLE TIME

= 60t/11 ± 180/11 minutes

or

= 60t/11 ± 15 × 12/11 minutes.

---

### Example: Between 4 and 5

First find the overlap time.

Overlap time:

= 60 × 4 / 11

= 240/11

= 21 9/11 minutes past 4.

### First right angle

Subtract:

180/11

Therefore,

= 240/11 − 180/11

= 60/11

= 5 5/11 minutes.

So,

First time = 4:05 5/11

### Second right angle

Add:

= 240/11 + 180/11

= 420/11

= 38 2/11 minutes.

So,

Second time = 4:38 2/11

Therefore,

ANSWERS:

4:05 5/11

and

4:38 2/11

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## 17. EXACT TIME WHEN THE TWO HANDS ARE OPPOSITE

When the hands are opposite, the angle between them is:

180°

From the overlap position, the minute hand must gain:

180°

Relative speed:

= 11/2° per minute

Time required:

= 180/(11/2)

= 360/11

= 32 8/11 minutes.

Therefore,

OPPOSITE TIME

= 60t/11 ± 360/11 minutes.

Only the value falling between t and t+1 is selected.

---

## SOLVED EXAMPLE 1

### Question

A watch which gains uniformly is 5 minutes slow at 8:00 a.m. on Sunday and it is 5 minutes 48 seconds fast at 8:00 p.m. on the following Sunday. When was it correct?

Options:

1. 6:00 p.m. on Wednesday
2. 7:20 p.m. on Wednesday
3. 6:00 p.m. on Tuesday
4. 7:20 p.m. on Tuesday

---

### Solution

At the beginning, the watch is:

5 minutes slow.

Therefore,

Error = −5 minutes.

At the end, it is:

5 minutes 48 seconds fast.

Convert 48 seconds into minutes:

48/60 = 4/5 minute.

Therefore,

5 min 48 sec

= 5 4/5 min

= 29/5 min.

Total change in error:

= 5 + 5 4/5

= 10 4/5 minutes

= 54/5 minutes.

---

### Step 1: Find total elapsed time

From 8:00 a.m. Sunday to 8:00 p.m. following Sunday:

7 days = 168 hours

Additional time = 12 hours

Total:

= 168 + 12

= 180 hours.

---

### Step 2: Find time required for watch to gain 5 minutes

The watch changes from 5 minutes slow to correct.

Therefore, it needs to gain:

5 minutes.

It gains 54/5 minutes in 180 hours.

Therefore,

Time required to gain 5 minutes:

= 180 × 5/(54/5)

= 180 × 5 × 5/54

= 180 × 25/54

= 3.333... × 25

= 83 1/3 hours

= 83 hours 20 minutes.

---

#### Step 3: Add this time to Sunday 8:00 a.m.

83 hours 20 minutes

= 3 days 11 hours 20 minutes.

Sunday 8:00 a.m. + 3 days

= Wednesday 8:00 a.m.

Add 11 hours 20 minutes:

= Wednesday 7:20 p.m.

Therefore,

ANSWER = 7:20 p.m. on Wednesday.

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## SOLVED EXAMPLE 2

### Question

A clock is set right at 1:00 p.m. If it gains one minute every hour, what is the true time when the clock indicates 6:00 p.m. on the same day?

Options:

1. 50 minutes past 5
2. 55 minutes past 5
3. 55 5/61 minutes past 5
4. 50 5/61 minutes past 5

---

### Solution

Clock indicates:

6:00 p.m.

Starting time:

1:00 p.m.

Therefore, according to the incorrect clock:

Elapsed time = 5 hours.

The clock gains 1 minute every hour.

Therefore, in 1 hour of clock time, actual time is:

60/61 hour.

Why?

In 60 minutes of true time, the clock shows:

61 minutes.

Therefore,

True time / Clock time

= 60/61.

---

### Step 1: Calculate actual elapsed time

Clock elapsed time = 5 hours.

True elapsed time:

= 5 × 60/61 hours

= 300/61 hours.

Convert into hours and minutes:

300/61

= 4 + 56/61 hours.

Therefore,

True elapsed time:

= 4 hours 56/61 hour.

Convert 56/61 hour into minutes:

= 56/61 × 60

= 3360/61

= 55 5/61 minutes.

Therefore,

True elapsed time:

= 4 hours 55 5/61 minutes.

---

#### Step 2: Add to 1:00 p.m.

1:00 p.m.

- 4 hours

= 5:00 p.m.

- 55 5/61 minutes

= 5:55 5/61 p.m.

Therefore,

ANSWER = 55 5/61 minutes past 5.

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## SOLVED EXAMPLE 3

### Question

A clock gains 5 minutes in 24 hours. It was set right at 10:00 a.m. on Monday. What will be the true time when the clock indicates 10:30 a.m. on the next Sunday?

Options:

1. 10:00 a.m. Sunday
2. 9:00 a.m. Sunday
3. 11:00 a.m. Sunday
4. None of these

---

### Solution

From Monday 10:00 a.m. to Sunday 10:30 a.m.:

Monday 10 a.m. to Sunday 10 a.m.

= 6 days

= 144 hours.

Additional:

= 30 minutes

= 1/2 hour.

Therefore,

Total clock time:

= 144 1/2 hours.

---

The clock gains:

5 minutes in 24 hours.

Therefore,

In 24 hours of incorrect clock time, it shows 24 hours 5 minutes.

Convert 5 minutes into hours:

5/60 = 1/12 hour.

Therefore,

24 hours of correct time corresponds to:

24 1/12 hours of clock time.

---

Let true elapsed time be x.

Then:

24 1/12 hours of clock time corresponds to 24 hours true time.

144 1/2 hours of clock time corresponds to x hours true time.

Therefore,

x/(144 1/2)

= 24/(24 1/12)

Hence,

x

= (144 1/2 × 24)/(24 1/12)

= 144 hours.

Therefore, true elapsed time = 144 hours = 6 days.

Starting time:

Monday 10:00 a.m.

After 6 days:

Sunday 10:00 a.m.

Therefore,

ANSWER = 10:00 a.m. on Sunday.

---

## SOLVED EXAMPLE 4

### Question

At what time between 3 and 4 o'clock are the hands of a clock together?

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### Solution

At 3:00:

Hour hand is at 3.

Minute hand is at 12.

Distance between them:

3 hour-spaces.

Each hour-space corresponds to 5 minute-spaces.

Therefore,

Gap:

= 3 × 5

= 15 minute-spaces.

---

The minute hand gains 55 minute-spaces over the hour hand in 60 minutes.

Therefore,

Time required to gain 15 spaces:

= 60/55 × 15

= 900/55

= 180/11

= 16 4/11 minutes.

Therefore,

The hands coincide:

16 4/11 minutes after 3.

ANSWER = 3:16 4/11.

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## SOLVED EXAMPLE 5

## Question

Find the time between 8 and 9 o'clock when the hands are in the same straight line but not together.

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## Solution

"Same straight line but not together" means:

Angle = 180°.

At 8:00:

Hour hand is at 8.

Minute hand is at 12.

Distance:

8 hour-spaces from 12.

But the smaller distance is:

12 − 8 = 4 hour-spaces.

Each hour-space = 5 minute-spaces.

Therefore,

Gap = 4 × 5

= 20 minute-spaces.

For opposite position, they must be 30 minute-spaces apart.

Therefore, additional gain required:

= 30 − 20

= 10 minute-spaces.

Minute hand gains 55 spaces in 60 minutes.

Therefore,

Time required:

= 60/55 × 10

= 600/55

= 120/11

= 10 10/11 minutes.

Therefore,

ANSWER = 8:10 10/11.

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## SOLVED EXAMPLE 6

### Question

If a clock takes 7 seconds to strike 7, how much time will it take to strike 10?

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### Solution

When a clock strikes 7:

Number of strikes = 7.

Number of intervals between strikes:

= 7 − 1

= 6 intervals.

Therefore,

6 intervals take 7 seconds.

---

When the clock strikes 10:

Number of intervals:

= 10 − 1

= 9 intervals.

Time for 9 intervals:

= 7/6 × 9

= 63/6

= 10.5 seconds.

Therefore,

ANSWER = 10 1/2 seconds.

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## SOLVED EXAMPLE 7

### Question

At what time between 5 and 6 are the minute and hour hands 7 minutes apart?

Here "7 minutes apart" means 7 minute-spaces apart.

There are two possible positions.

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## Case 1: Minute hand is behind the hour hand

At 5:00:

Hour hand is at 5.

Minute hand is at 12.

Gap:

= 5 × 5

= 25 minute-spaces.

To become 7 spaces apart:

Required reduction:

= 25 − 7

= 18 spaces.

Therefore, minute hand must gain 18 spaces.

Time:

= 60/55 × 18

= 1080/55

= 216/11

= 19 7/11 minutes.

Therefore,

First time:

5:19 7/11.

---

## Case 2: Minute hand is ahead of the hour hand

The minute hand must gain enough to move from 25 spaces behind to 7 spaces ahead.

Required gain:

= 25 + 7

= 32 spaces.

Time:

= 60/55 × 32

= 1920/55

= 384/11

= 34 10/11 minutes.

Therefore,

Second time:

5:34 10/11.

### ANSWER

The hands are 7 minutes apart at:

1. 5:19 7/11
2. 5:34 10/11

---

## SOLVED EXAMPLE 8

### Question

A minute hand of a clock overtakes the hour hand at intervals of 63 minutes of correct time. How much time does the clock gain or lose per day?

Options:

1. 55 8/76 min
2. 56 8/77 min
3. 56 9/77 min
4. 77 8/56 min

---

### Solution

In a correct clock, the minute hand overtakes the hour hand once every:

60/11 hours

Convert to minutes:

= 60/11 × 60

= 3600/11

= 327 3/11 minutes?

This is not the relevant interval when considering successive overlaps on the clock face. The standard interval between successive overlaps is:

65 5/11 minutes.

This is:

65 5/11 minutes.

But the given clock has an overlap interval of only:

63 minutes of correct time.

Therefore, the clock is running fast.

---

### Step 1: Find gain during 63 minutes

Correct interval:

65 5/11 minutes

Actual interval:

63 minutes

Gain:

= 65 5/11 − 63

= 2 5/11

= 27/11 minutes.

Therefore, the clock gains:

27/11 minutes in every 63 minutes of correct time.

---

### Step 2: Find gain in one day

One day:

= 24 hours

= 24 × 60

= 1440 minutes.

Number of 63-minute intervals in one day:

= 1440/63.

Therefore, gain:

= 27/11 × 1440/63

Simplify:

= (27 × 1440)/(11 × 63)

Since 27/63 = 3/7,

= (3 × 1440)/(11 × 7)

= 4320/77

= 56 8/77 minutes.

Therefore,

ANSWER = 56 8/77 minutes gain per day.

---

## SOLVED EXAMPLE 9

### Question

At what time between 4 and 5 will the hands of a clock be at right angles?

---

### Solution

At 4:00:

Hour hand is at 4.

Minute hand is at 12.

Difference:

= 4 × 5

= 20 minute-spaces.

For right angle:

90° = 15 minute-spaces.

---

### Case 1

Minute hand is behind the hour hand.

It needs to reduce the gap from 20 spaces to 15 spaces.

Required gain:

= 20 − 15

= 5 spaces.

Time:

= 60/55 × 5

= 300/55

= 60/11

= 5 5/11 minutes.

Therefore,

Time = 4:05 5/11.

---

### Case 2

Minute hand passes the hour hand and becomes 15 spaces ahead.

Required gain:

= 20 + 15

= 35 spaces.

Time:

= 60/55 × 35

= 2100/55

= 420/11

= 38 2/11 minutes.

Therefore,

Time = 4:38 2/11.

#### ANSWER

4:05 5/11 and 4:38 2/11.

---

## SOLVED EXAMPLE 10

### Question

At what time between 5 and 6 will the minute and hour hands be 3 minutes apart?

---

## Solution

At 5:00:

Hour hand is at 5.

Minute hand is at 12.

Gap:

= 5 × 5

= 25 spaces.

Required difference = 3 spaces.

---

## Case 1: Minute hand behind

It must reduce the gap from 25 spaces to 3 spaces.

Required gain:

= 25 − 3

= 22 spaces.

Time:

= 60/55 × 22

= 24 minutes.

Therefore,

First time = 5:24.

---

## Case 2: Minute hand ahead

It must gain:

= 25 + 3

= 28 spaces.

Time:

= 60/55 × 28

= 1680/55

= 336/11

= 30 6/11 minutes.

Therefore,

Second time:

= 5:30 6/11.

#### ANSWER

5:24 and 5:30 6/11.

---

## SOLVED EXAMPLE 11

### Question

A minute hand of a clock overtakes the hour hand at intervals of 65 minutes of correct time. How much does the clock gain or lose per day?

---

### Solution

In a correct clock, successive overlap occurs every:

65 5/11 minutes.

Given clock:

65 minutes.

Therefore, the clock completes the cycle faster.

Hence, the clock is **gaining**.

Gain in one cycle:

= 65 5/11 − 65

= 5/11 minute.

Therefore,

Gain = 5/11 minute in 65 minutes.

---

#### Gain in 24 hours

24 hours = 1440 minutes.

Gain:

= 5/11 × 1440/65

= 7200/715

= 1440/143

= 10 10/43 minutes.

Therefore,

ANSWER = 10 10/43 minutes gain per day.

---

## SOLVED EXAMPLE 12

### Question

A watch gains uniformly. It is 5 minutes slow at 8:00 a.m. Sunday and 5 minutes 48 seconds fast at 8:00 p.m. the following Sunday. Find when it was correct.

This is the same principle as Example 1, but another useful method is:

Initial error = −5 minutes

Final error = +5 4/5 minutes

Total change in error:

= 10 4/5

= 54/5 minutes.

Total time:

= 180 hours.

Rate of correction:

= 54/5 ÷ 180

= 54/(5 × 180)

= 3/50 minute per hour.

Time required to correct 5 minutes:

= 5 ÷ 3/50

= 5 × 50/3

= 250/3

= 83 1/3 hours

= 83 hours 20 minutes.

Therefore:

83 hours 20 minutes after Sunday 8:00 a.m.

= Wednesday 7:20 p.m.

ANSWER = 7:20 p.m. Wednesday.

---

## VERY IMPORTANT SHORTCUTS FOR EXAM

### 1. Angle Formula

At H:M:

ANGLE = |30H − 11M/2|

Remember:

Hour hand position = 30H + M/2

Minute hand position = 6M

---

### 2. Relative Speed

Minute hand:

6°/min

Hour hand:

1/2°/min

Therefore:

Relative speed = 6 − 1/2

= 11/2°/min.

This is the most important value in clock problems.

---

### 3. Coincidence

Between t and t+1:

Time = 60t/11 minutes past t.

---

### 4. Right Angle

Time:

= 60t/11 ± 180/11 minutes.

---

## 5. Opposite Hands

Time:

= 60t/11 ± 360/11 minutes.

---

## 6. Minute-Spaces Method

At exactly t:00:

Gap between hands

= 5t minute-spaces.

For a required separation of x spaces:

### If minute hand is behind:

Required gain = 5t − x

### If minute hand is ahead:

Required gain = 5t + x

Then:

Time = (60/55) × Required gain.

This method is especially useful for questions such as:

- hands are 3 minutes apart
- hands are 5 minutes apart
- hands are 7 minutes apart
- hands are at right angles
- hands are in a straight line
- hands coincide.

---

## 7. Clock Gain/Loss

If a clock gains:

x minutes in y minutes,

then gain per day:

= x/y × 1440 minutes.

If a clock loses:

x minutes in y minutes,

then loss per day:

= x/y × 1440 minutes.

For questions involving two displayed times, use:

**True Time / Clock Time = Correct Rate / Incorrect Rate**

---

## 8. Striking Clock Problems

If a clock takes T seconds to strike n times:

Number of intervals = n − 1.

Therefore,

Time for one interval:

= T/(n − 1).

For m strikes:

Time required:

= T/(n − 1) × (m − 1).

Example:

7 strikes → 6 intervals

10 strikes → 9 intervals

Therefore:

Time for 10 strikes

= 7/6 × 9

= 10.5 seconds.